

AI ASSISTED LAB ASSIGNMENT-6.1

NAME: N. Adithya

BATCH:14

ROLL_NO:2503A52413

Task-1: (AI-Based Code Completion for Loops)

Prompt:

“Generate Python code to print all even numbers between 1 and N

Using a loop.”

Input:

#Generate Python code to print all even numbers between 1 and N using a loop.

```
N = int(input("Enter a number: "))
```

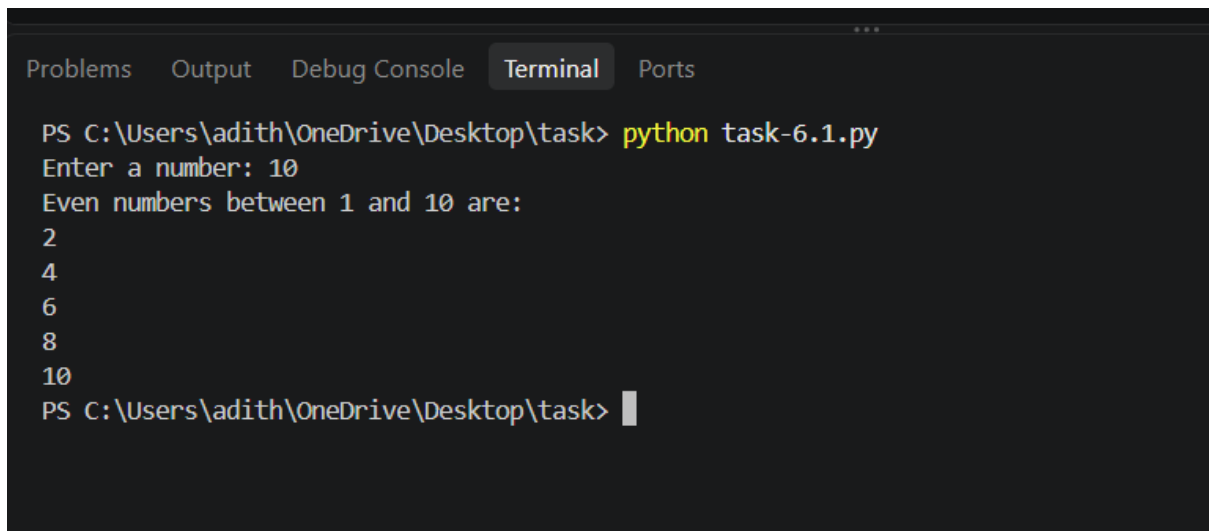
```
print("Even numbers between 1 and", N, "are:")
```

```
for i in range(1, N + 1):
```

```
    if i % 2 == 0:
```

```
        print(i)
```

output:

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'Problems', 'Output', 'Debug Console', 'Terminal' (which is selected), and 'Ports'. The terminal shows the command prompt 'PS C:\Users\adith\OneDrive\Desktop\task>' followed by the command 'python task-6.1.py'. The program then prompts 'Enter a number: 10'. It then outputs 'Even numbers between 1 and 10 are:' followed by a list of even numbers: '2', '4', '6', '8', and '10'. Finally, it returns to the command prompt 'PS C:\Users\adith\OneDrive\Desktop\task>' with a cursor.

```
PS C:\Users\adith\OneDrive\Desktop\task> python task-6.1.py
Enter a number: 10
Even numbers between 1 and 10 are:
2
4
6
8
10
PS C:\Users\adith\OneDrive\Desktop\task> 
```

Task-2: (AI-Based Code Completion for Loop withConditionals)

Prompt:

“Generate Python code to count how many numbers in a list are even and odd.”

Input:

```
def count_even_odd(numbers):
    even_count = 0
    odd_count = 0

    for num in numbers:
        if num % 2 == 0:
            even_count += 1
        else:
            odd_count += 1
```

```
return even_count, odd_count
```

```
# Example usage
```

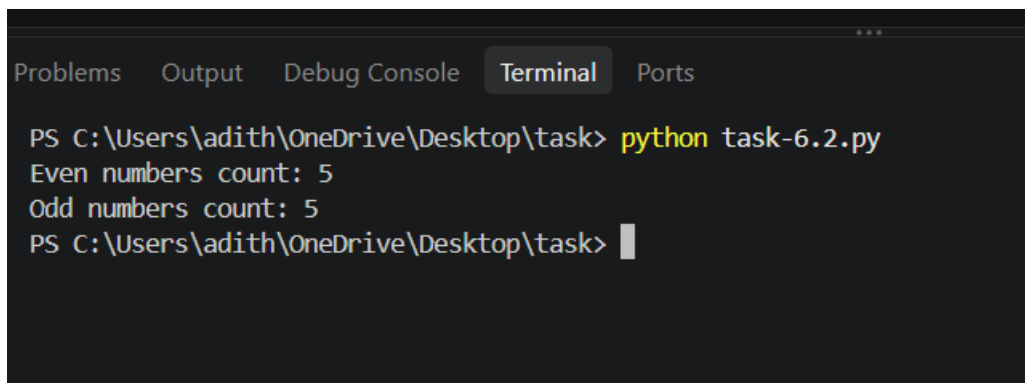
```
numbers_list = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

```
even_count, odd_count = count_even_odd(numbers_list)
```

```
print(f"Even numbers count: {even_count}")
```

```
print(f"Odd numbers count: {odd_count}")
```

output:

A screenshot of a Windows terminal window with a dark background. The terminal has tabs at the top labeled 'Problems', 'Output', 'Debug Console', 'Terminal' (which is active), and 'Ports'. The command prompt shows the user running 'python task-6.2.py' in the directory 'C:\Users\adith\OneDrive\Desktop\task'. The output of the script is displayed on the next two lines: 'Even numbers count: 5' and 'Odd numbers count: 5'. The prompt then returns to 'PS C:\Users\adith\OneDrive\Desktop\task>' with a cursor.

Task-3: (AI-Based Code Completion for Class Attributes Validation)

Prompt:

“Generate a Python class User that validates age and email using conditional statements.”

Input:

```
class User:
```

```
def __init__(self, name, age, email):
```

```
    self.name = name
```

```
    self.age = age
```

```
    self.email = email
```

```
def validate_age(self):
```

```
    if self.age < 0:
```

```
        return "Age cannot be negative."
```

```
    elif self.age < 18:
```

```
        return "User is a minor."
```

```
    else:
```

```
        return "User is an adult."
```

```
def validate_email(self):
```

```
    if "@" not in self.email or "." not in self.email:
```

```
        return "Invalid email format."
```

```
    else:
```

```
        return "Email is valid."
```

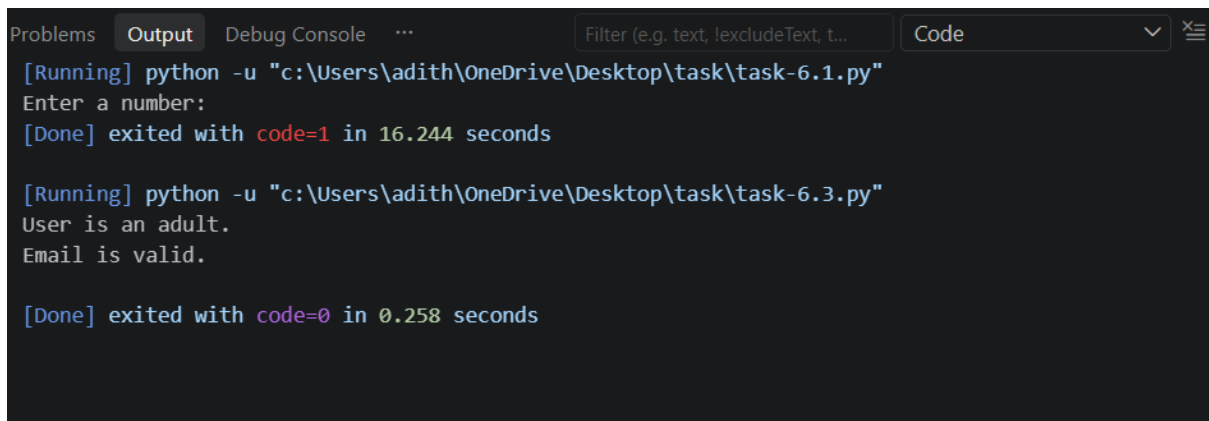
```
# Example usage:
```

```
user1 = User("Alice", 25, "alice@example.com")
```

```
print(user1.validate_age()) # Output: User is an adult.
```

```
print(user1.validate_email()) # Output: Email is valid.
```

Output:



```
Problems Output Debug Console ... Filter (e.g. text, excludeText, t...) Code
[Running] python -u "c:\Users\adith\OneDrive\Desktop\task\task-6.1.py"
Enter a number:
[Done] exited with code=1 in 16.244 seconds

[Running] python -u "c:\Users\adith\OneDrive\Desktop\task\task-6.3.py"
User is an adult.
Email is valid.

[Done] exited with code=0 in 0.258 seconds
```

Task-4: (AI-Based Code Completion for Classes)

Prompt:

“Generate a Python class Student with attributes (name, roll number, marks) and methods to calculate total and average marks.”

Input:

```
class Student:

    def __init__(self, name, roll_number, marks):

        self.name = name

        self.roll_number = roll_number

        self.marks = marks

    def calculate_total(self):

        return sum(self.marks)

    def calculate_average(self):

        total = self.calculate_total()
```

```
return total / len(self.marks) if self.marks else 0
```

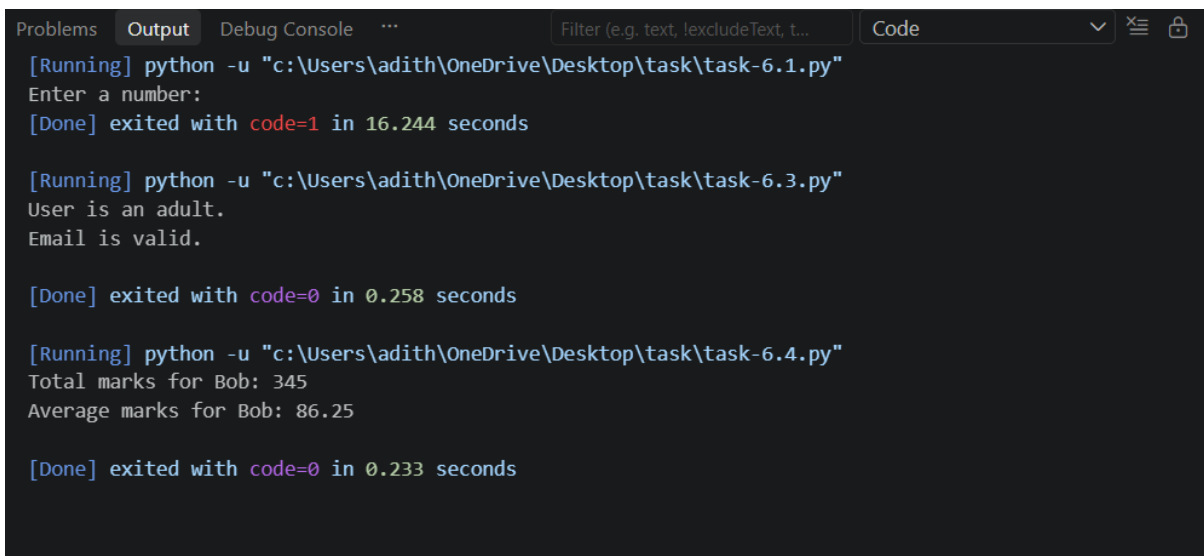
Example usage:

```
student1 = Student("Bob", 101, [85, 90, 78, 92])
```

```
print(f"Total marks for {student1.name}: {student1.calculate_total()}")
```

```
print(f"Average marks for {student1.name}: {student1.calculate_average()}")
```

output:



```
Problems Output Debug Console ... Filter (e.g. text, excludeText, t...) Code
[Running] python -u "c:\Users\adith\OneDrive\Desktop\task\task-6.1.py"
Enter a number:
[Done] exited with code=1 in 16.244 seconds

[Running] python -u "c:\Users\adith\OneDrive\Desktop\task\task-6.3.py"
User is an adult.
Email is valid.
[Done] exited with code=0 in 0.258 seconds

[Running] python -u "c:\Users\adith\OneDrive\Desktop\task\task-6.4.py"
Total marks for Bob: 345
Average marks for Bob: 86.25
[Done] exited with code=0 in 0.233 seconds
```

Task-5: (AI-Assisted Code Completion Review)

Prompt:

“Generate a Python program for a simple bank account system using class, loops, and conditional statements.”

Input:

#Generate a Python program for a simple bank account system using class, loops, and conditional statements.

```
class BankAccount:
```

```
    def __init__(self, account_number, account_holder, balance=0):
```

```
self.account_number = account_number

self.account_holder = account_holder

self.balance = balance


def deposit(self, amount):

    if amount > 0:

        self.balance += amount

        print(f"Deposited: ${amount}. New balance: ${self.balance}.")

    else:

        print("Deposit amount must be positive.")


def withdraw(self, amount):

    if amount > 0:

        if amount <= self.balance:

            self.balance -= amount

            print(f"Withdrew: ${amount}. New balance: ${self.balance}.")

        else:

            print("Insufficient funds.")

    else:

        print("Withdrawal amount must be positive.")


def check_balance(self):

    print(f"Account Balance: ${self.balance}.")

    #give example usage

if __name__ == "__main__":

    # Create a new bank account

    account = BankAccount("12345", "John Doe", 1000)
```

```
# Check initial balance

account.check_balance()


# Deposit money

account.deposit(500)

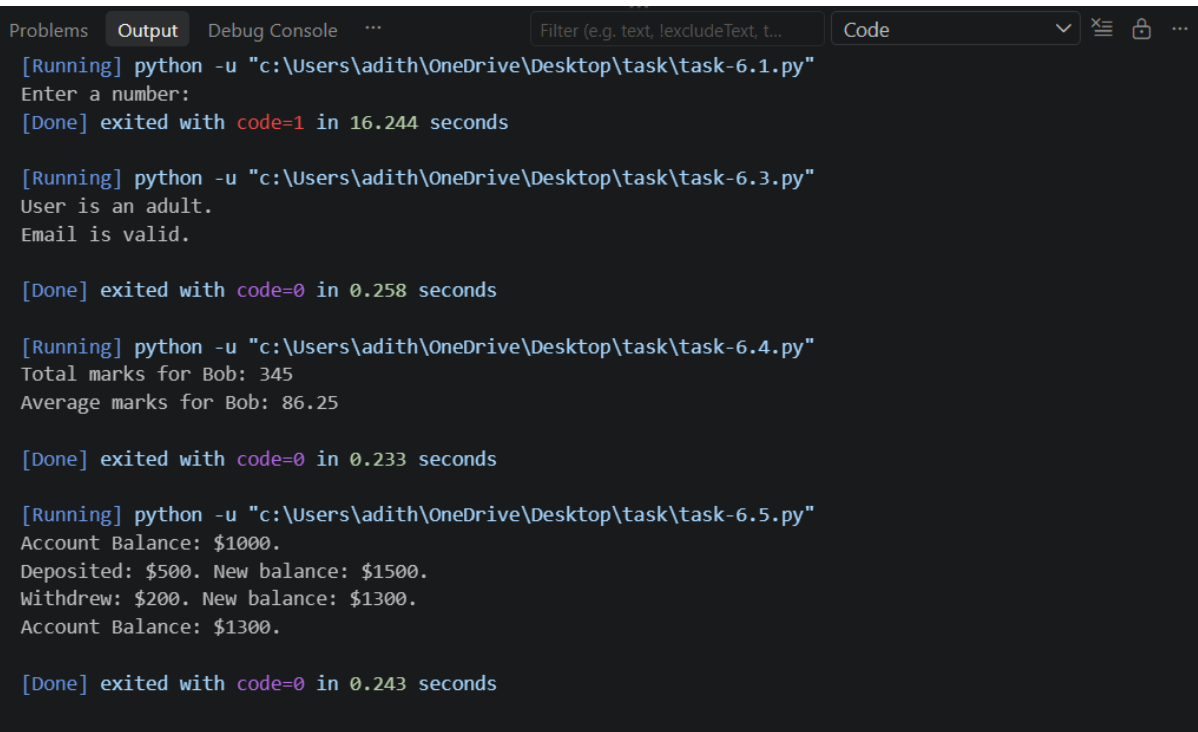

# Withdraw money

account.withdraw(200)


# Check final balance

account.check_balance()
```

Output:



```
Problems Output Debug Console ... Filter (e.g. text, excludeText, t...) Code
[Running] python -u "c:\Users\adith\OneDrive\Desktop\task\task-6.1.py"
Enter a number:
[Done] exited with code=1 in 16.244 seconds

[Running] python -u "c:\Users\adith\OneDrive\Desktop\task\task-6.3.py"
User is an adult.
Email is valid.
[Done] exited with code=0 in 0.258 seconds

[Running] python -u "c:\Users\adith\OneDrive\Desktop\task\task-6.4.py"
Total marks for Bob: 345
Average marks for Bob: 86.25
[Done] exited with code=0 in 0.233 seconds

[Running] python -u "c:\Users\adith\OneDrive\Desktop\task\task-6.5.py"
Account Balance: $1000.
Deposited: $500. New balance: $1500.
Withdrew: $200. New balance: $1300.
Account Balance: $1300.
[Done] exited with code=0 in 0.243 seconds
```